

Multicenter Prospective Study in HER2-Positive Early Breast Cancer for Detecting Minimal Residual Disease by Circulating Tumor DNA Analysis With Neoadjuvant Chemotherapy: HARMONY Study

Breast Cancer: Basic and Clinical Research
Volume 18: 1–13
© The Author(s) 2024
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/11782234241288671



Momoko Tokura^{1,2,3} , Mark Malalay Ando⁴, Yuki Kojima^{1,2}, Rui Kitadai¹, Shu Yazaki^{1,2}, Cyrielle Marie N Atutubo⁵, Rubi K. Li⁵, Minda Z. Perez⁶, Agnes E Gorospe⁵, Manuelito A Madrid⁷, Mel Valerie C Ordinario⁵, Marcelo Severino B Imasa⁵, Kazuki Sudo^{1,2,8}, Tatsunori Shimoi^{1,2}, Akihiko Suto⁹, Shinji Kohsaka¹⁰, Ryunosuke Machida¹¹, Ryo Sadachi¹¹, Masayuki Yoshida¹², Yasushi Yatabe¹², Tomomi Hata¹³, Kenichi Nakamura², Kan Yonemori^{1,2,8} and Sho Shiino⁹

¹Department of Medical Oncology, National Cancer Center Hospital, Tokyo, Japan. ²Department of International Clinical Development, National Cancer Center Hospital, Tokyo, Japan.

³Laboratory of Integrative Oncology, National Cancer Center Research Institute, Tokyo, Japan.

⁴Department of Medical Oncology, Cebu South Medical Center, Talisay, Philippines. ⁵Section of Medical Oncology, St. Luke's Medical Center, Quezon, Philippines. ⁶Institution of Surgery, St. Luke's Medical Center, Quezon, Philippines. ⁷Department of Pathology, St. Luke's Medical Center, Quezon, Philippines. ⁸Department of Experimental Therapeutics, National Cancer Center Hospital, Tokyo, Japan. ⁹Department of Breast Surgery, National Cancer Center Hospital, Tokyo, Japan. ¹⁰Division of Cellular Signaling, National Cancer Center Research Institute, Tokyo, Japan.

¹¹Biostatistics Section, Clinical Research Support Office, National Cancer Center Hospital, Tokyo, Japan. ¹²Department of Diagnostic Pathology, National Cancer Center Hospital, Tokyo, Japan.

¹³International Trial Management Section, Research Management Division, Clinical Research Support Office, National Cancer Center Hospital, Tokyo, Japan.

ABSTRACT

BACKGROUND: Biomarkers to predict the recurrence risk are required to optimize perioperative treatment. Adjuvant chemotherapy for patients with human epidermal growth factor 2-positive (HER2-positive) early breast cancer is decided by pathological responses of neoadjuvant chemotherapy (NAC). However, whether pathological responses are appropriate biomarkers is unclear. Currently, there are several studies using minimal residual disease (MRD) as a predictor of prognosis in solid tumors. However, there is no standard method for detecting MRD.

OBJECTIVES: This study aimed at prospectively evaluating the relationship between MRD detection and recurrence in Asian patients with HER2-positive early breast cancer.

DESIGN: Prospective, observational, single-group, and exploratory. This study will include 60 patients from 2 institutions in Japan and the Philippines. The invasive disease-free survival (IDFS) rates of the MRD-positive and MRD-negative groups are compared in patients with HER2-positive early breast cancer who undergo surgery after receiving NAC.

METHODS AND ANALYSIS: Circulating tumor DNA (ctDNA) levels of patients will be evaluated 6 times: before NAC, after NAC, after surgery, and annually after surgery for 3 years. We will analyze the genetic profile of blood and tissue samples using the Todai OncoPanel (TOP) and the methylation level of DNA. The primary endpoint is IDFS. Secondary endpoints include overall survival (OS) and disease-free survival (DFS). Patient enrollment began in June 2022, and new participants are still being recruited.

ETHICS: This study has been approved by the National Cancer Center Hospital Certified Review Board in March 2022 and has been approved by the Research Ethics Board of the participating center.

DISCUSSION: Our findings will contribute to determining whether MRD detection using TOP is useful for predicting the recurrence of HER2-positive early breast cancer. If this is proven, MRD detected by TOP could be used in the future as a biomarker to assist in the de/escalation of treatment strategies in the next interventional trial, thereby avoiding overtreatment in patients at low risk, and in the addition of intensive treatment modalities for those in patients at high risk.

KEYWORDS: HER2-positive breast cancer, minimal residual disease, ctDNA, ctRNA, Todai OncoPanel, Asia

RECEIVED: January 25, 2024. **ACCEPTED:** September 5, 2024.

TYPE: Study Protocol

FUNDING: The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This study is supported by AMED (Japan Agency for Medical Research and Development) (grant nos. 20lk0201002j0001 and 21lk0201005j0001). This funding source had no role in the design of this study and will not have any role during its execution, analyses, interpretation of the data, or decision to submit results.

COMPETING INTERESTS: The author(s) declared the following potential conflicts of interest with respect to the research, authorship, and/or publication of this article: Part of this research will be conducted under a joint research agreement between the National Cancer Center and the H.U. Group Research Institute G.K., a limited liability company that will bear the costs associated with specimen processing.

CORRESPONDING AUTHORS: Yuki Kojima, Department of Medical Oncology, National Cancer Center Hospital, 5-1-1 Tsukiji, Chuo-ku, Tokyo 104-0045, Japan. Email: yuukojim@ncc.go.jp

Sho Shiino, Department of Breast Surgery, National Cancer Center Hospital, 5-1-1 Tsukiji, Chuo-ku, Tokyo 104-0045, Japan. Email: sshiino@ncc.go.jp



Creative Commons Non Commercial CC BY-NC: This article is distributed under the terms of the Creative Commons Attribution-NonCommercial 4.0 License (<https://creativecommons.org/licenses/by-nc/4.0/>) which permits non-commercial use, reproduction and distribution of the work without further permission provided the original work is attributed as specified on the SAGE and Open Access pages (<https://us.sagepub.com/en-us/nam/open-access-at-sage>).

Introduction

Globally, the incidence of breast cancer in 2020 was estimated at 2.3 million cases, with half of these deaths occurring in Asia.¹ The number of breast cancer survivors is increasing because of the rise in the incidence of breast cancer and improvements in treatment strategies for breast cancer.^{2,3} However, the mortality rates remain high in Asian countries. The 5-year survival rate of patients with breast cancer in Asia is 65% to 89%.⁴ Perioperative treatment strategies are important to reduce mortality in breast cancer survivors but biomarkers to determine appropriate adjuvant therapy have not been explored.

Human epidermal growth factor 2-positive (HER2-positive) breast cancer accounts for 15% to 20% of breast cancers and is common in Asia.⁵ Neoadjuvant chemotherapy (NAC) with anti-HER2 drugs has dramatically improved pathological complete response (pCR) rate, and prognosis and survivors are increasing.⁶ Pathological evaluation of treatment effect is often linked to prognosis, and more intensive treatments, such as adjuvant trastuzumab emtansine (T-DM1), are recommended for patients with HER2-positive breast cancer who cannot achieve pCR after NAC.⁷ However, the recurrence rate of invasive disease in patients with HER2-positive breast cancer who achieved pCR after dual HER2 blockade could still be as high as 10%, with higher rates among those with node-positive disease.⁸ This is partly because of recurrence, which can be attributed to micrometastatic disease at the initial diagnosis that is not detected by preliminary diagnostics.^{9,10} Neoadjuvant chemotherapy aims to eradicate these micrometastases, translating to increased pCR and a substantial decrease in relapse risk.¹¹ Although pCR is often linked to favorable outcomes and low recurrence rates, better diagnostic tools for residual disease are needed. There are some genetic assays for predicting the effect of chemotherapy and prognosis. Oncotype DX is a multigene assay for predicting chemotherapy treatment effects and prognosis in patients with hormonal receptor-positive and HER2-negative breast cancer without NAC.^{12,13} HER2DX is a test to estimate the risk of recurrence and pCR likelihood in patients with HER2-positive early breast cancer. This test incorporates tumor size, nodal staging, and 4 gene expression signatures tracking immune infiltration, tumor cell proliferation, luminal differentiation, and the expression of HER2 amplicon into a single score.¹⁴ However, these tests do not reflect the status of residual tumor cells after treatment completion.

In recent years, detecting circulating tumor DNA (ctDNA) in patients with early cancer has been used for tracking minimal residual disease (MRD).^{15,16} Both ctDNA analysis after completion of all treatments and the surveillance ctDNA analysis are associated with recurrence in breast cancer.¹⁷⁻¹⁹ However, there are no standard methods of ctDNA detection, and methods differ depending on the study.¹⁵ Circulating

tumor DNA detection in patients with early-stage cancer is challenging due to the low tumor burden.^{15,16} Therefore, a study on ctDNA detection methods with sufficient sensitivity and specificity for clinical use is needed.

In this study, we plan to detect ctDNA using the Todai OncoPanel (TOP) and analysis of DNA methylation level. We will prospectively evaluate the association between ctDNA detected by this method and prognosis in patients with HER2-positive breast cancer who received NAC in Japan and the Philippines. If ctDNA can be detected correctly, ctDNA-positive patients will have a higher recurrence rate, and ctDNA-negative patients should show a lower recurrence rate.

Methods/design

Aim

This study aims to evaluate the relationship between MRD detection and recurrence in patients with HER2-positive early-stage breast cancer. In addition, we aim to establish a new MRD detection assay for HER2-positive breast cancer.

Study setting

This is a prospective, observational, single-group, exploratory, and multicenter study (National Cancer Center Hospital in Japan and St. Luke's Medical Center in the Philippines). This study includes cases of HER2-positive breast cancer before NAC followed by surgery, so that the site must be a hospital with breast cancer surgery and a medical oncology department. We chose a participating site with sufficient cases and availability of collection blood and tissue samples with sufficient quality to extract nucleic acid. We summarize study registration data in Table 1 and an overview of this study in Figure 1. Regimens of adjuvant chemotherapy for non-pCR cases are determined by each institution. No financial or nonfinancial incentives will be provided in this study.

Endpoints

The primary endpoint is invasive disease-free survival (IDFS). Invasive disease-free survival is defined as the duration from the date of surgery to the date of the judgment of one of the following events, whichever occurs first: ipsilateral and contralateral invasive breast cancer recurrence, regional invasive breast cancer recurrence, distant recurrence, and death attributable to any cause.

The secondary endpoints are overall survival (OS) and disease-free survival (DFS). Overall survival is defined as the duration from the date of surgery to the death attributable to any cause. Disease-free survival is defined as the duration from the date of surgery to the date of the judgment of one of the following events, whichever occurs first: IDFS events,

Table 1. Trial registration data.

DATA CATEGORY	INFORMATION
Trial identifying number	ClinicalTrials.gov: NCT05433753
Date of registration in primary registry	June 27, 2022
Secondary identifying numbers	NA
Source of monetary or material support	AMED (Japan Agency for Medical Research and Development)
Primary sponsor	National Cancer Center Hospital, Japan
Secondary sponsor	St. Luke's Medical Center, Philippines
Contact for public queries	Coordinating office (E-mail: NCCH2113_office@ml.res.ncc.go.jp)
Contact for scientific queries	Coordinating office (E-mail: NCCH2113_office@ml.res.ncc.go.jp)
Public title	Multicentre Prospective Study in HER2-Positive Early Breast Cancer for Detecting Minimal Residual Disease by Circulating Tumor DNA Analysis with Neoadjuvant Chemotherapy: HARMONY Study
Scientific title	Study in HER2-Positive Early Breast Cancer for Detecting Minimal Residual Disease by Circulating Tumor DNA Analysis with Neoadjuvant Chemotherapy: HARMONY Study-Multicentre, Prospective, Observational Study
Countries of recruitment	Japan, Philippines
Health condition or problem studied	MRD detection of early breast cancer patients with NAC
Intervention	NA
Key inclusion and exclusion criteria	Aged of 20 years or older at the date of registration. Inclusion criteria: histological diagnosis of HER2-positive invasive breast carcinoma, neoadjuvant chemotherapy followed by surgery is scheduled, clinical stages IIA-IIIC. Exclusion criteria: any other malignancy within 5 years before registration, bilateral synchronous breast cancer, history of breast cancer, pregnancy during registration
Study type	Prospective, observational Multicenter Exploratory study
Date of first enrolment	June 27, 2022
Target sample size	60
Recruitment status	Recruiting
Primary outcome	Invasive disease-free survival (IDFS)
Key secondary outcomes	Overall survival (OS), disease-free survival (DFS)

ipsilateral or contralateral ductal carcinoma in situ (DCIS), or second primary nonbreast invasive cancer.

Eligibility criteria

1. Age: 20 years or older at the date of registration.
2. Histological diagnosis of HER2-positive invasive breast carcinoma.
3. Scheduled for neoadjuvant chemotherapy followed by surgery.
4. Clinical stages IIA-IIIC (AJCC/UICC TNM staging system, seventh edition).

5. Known hormone receptor status.
6. Provision of signed informed consent.

Exclusion criteria

1. Any other malignancy within 5 years before registration, except for adequately treated basal cell or squamous cell skin cancer, or carcinoma in situ of the esophagus, stomach, colon, or cervix.
2. Bilateral synchronous breast cancer.
3. History of breast cancer.
4. Pregnancy during registration.

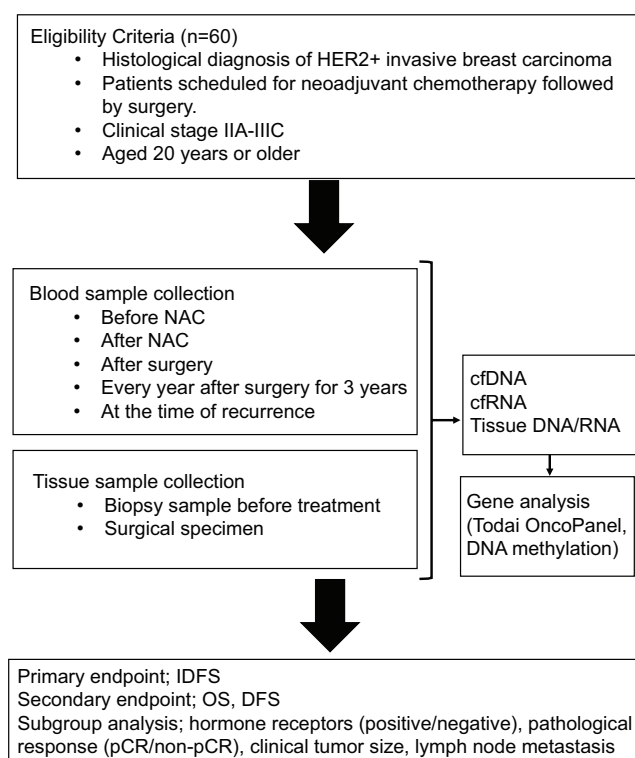


Figure 1. Flow diagram of the HARMONY protocol.

The eligibility criteria are HER2-positive invasive breast carcinoma, scheduled for neoadjuvant chemotherapy followed by surgery, and clinical stages IIA-IIIC. A total of 60 patients will be enrolled. Blood samples are collected 6 times: before neoadjuvant chemotherapy, after neoadjuvant chemotherapy, after surgery, and annually after surgery for 3 years. When recurrence occurs, blood is collected only at that time, and no further blood samples are collected. Biopsy samples before treatment and surgical specimens will be collected. Genetic analysis of blood and tissue samples will be performed for MRD detection. The primary endpoint is the invasive disease-free survival (IDFS), whereas the secondary endpoints are overall survival (OS) and disease-free survival (DFS). Subgroup analysis will be performed.

Sample size calculation and statistical methods

The target enrollment number is 60 patients. Because this is an exploratory study, the sample size is determined by considering feasibility. Each site performs NAC on more than 20 new patients with breast cancer each year, so it will be possible to enroll 60 patients more than a 3-year period. Patient enrollment began in June 2022, and the participant recruitment is currently incomplete.

For all eligible cases, the IDFS, median IDFS, and annual IDFS proportions in the MRD-positive and MRD-negative group are estimated using the Kaplan-Meier method. The 95% confidence intervals of the median IDFS and annual IDFS proportions are estimated using the Brookmeyer and Crowley method and the Greenwood formula, respectively. The same analysis is performed on the pooled population.

The IDFS rates of the MRD-positive and MRD-negative groups are compared using an unstratified log-rank test; however, this result should be interpreted as being exploratory, as the sample size is not based on any statistical tests in this exploratory study. For all eligible cases, the hazard ratio (HR) between the MRD-positive and MRD-negative groups and its

95% confidence interval is estimated using an unstratified Cox proportional hazards model.

The group excluding “ineligible patients” determined by principal investigator or Coordinating Investigators from all registered patients is defined as “all eligible patients.” Among all eligible patients who completed NAC and surgery, the group of patients who submitted the available blood sample collected within 28 days after surgery for MRD analysis is defined as “all eligible examination cases.” This is the primary data set, and we will perform statistical analysis for the following groups; all registered patients, all eligible patients, patients with NAC, patients who completed NAC, patients with surgery, patients who completed surgery, and all examination cases.

Factors planned for subgroup analysis are as follows:

- Age, years (<50/≥50)
- Race
- Clinical T factor (T1/T2/T3/T4)
- Clinical N factor (N0/N1/N2/N3)
- Menopausal status (pre/post)
- Family history (yes/no)
- Smoking history (yes/no)
- Hormone receptors (positive/negative) (biopsy tissue sample)
- Histological grade, low (grades 1 and 2)/high (grade 3) (biopsy tissue sample)
- Anti-HER2 drug (used/unused)
- Genetic abnormality before NAC
- Pathological response (pCR/non-pCR)

Statistical considerations

This study is designed to evaluate the relationship between recurrence and ctDNA positivity. The dropout rate is estimated at 10%, and the recurrence rate within 3 years is estimated at 20%. Owing to the small sample size, we plan to evaluate the results using point estimates. Individuals in the study group will not be matched to a control group.

Patient enrollment

A trained surgeon or medical oncologist will introduce the study to patients after diagnosis using an informed consent form (ICF). Patients who agree to participate in the study will be registered using the Electronic Data Capture (EDC) system.

Procedures for sample collection

Blood samples are collected 6 times: before NAC, after NAC, after surgery, and annually after surgery for 3 years. When recurrence occurs, a blood sample is collected only at that time, and no further blood samples are collected. The definition of

	Enrolment	Treatment period			Follow up period			
		Before NAC	After NAC	After surgery	One year after surgery	Two years after surgery	Three years after surgery	At the time of recurrence
Eligibility screen	•							
Informed consent	•							
Tissue sample collection		•		•				
Blood sample collection		•	•	•	•	•	•	•
Pathological findings and biomarker tests	•			•				
Recurrence status					•	•	•	•
Survival status					•	•	•	•

Figure 2. Schedule of the study.

Eligibility screening, pathological findings of biopsy sample, and informed consent are required before enrollment. Tissue samples and blood samples are collected during treatment period. Blood samples and information about recurrence/survival status are collected during follow-up period.

recurrence is IDFS events except “death attributable to any cause.” If the recurrence is DCIS, only an EDC report is required, and no blood sampling is needed. Tissue samples from the biopsy and surgery are collected for genetic analysis. The schedule of enrollment, sample collection, and assessments is shown in Figure 2. Plasma separation is performed at each site by a trained research assistant. Plasma, buffy coat, and tissue samples from each site will be collected at the National Cancer Center, and genomic analysis will be performed by the National Cancer Center Research Institute.

MRD detection methods

In this study, MRD is defined as ctDNA/ctRNA detected in the absence of clinical evidence of tumor after surgery. Cell-free DNA (cfDNA)/cell-free RNA (cfRNA) and buffy-coat DNA will be extracted from blood samples. Buffy-coat DNA will be used for filtering genomic DNA for assessing the clonal hematopoiesis of indeterminate potential (CHIP). Circulating tumor DNA will be detected by TOP and methylation analyses. Cell-free DNA with cancer-related genetic abnormalities will be monitored as ctDNA. We will extract cfRNA and apply TOP to identify ctRNA. As this is the first study of TOP for ctDNA detection in the MRD setting, its sensitivity and specificity for MRD detection will be validated herein. The minimum detection sensitivity of SNV/indel tested by the TOP using commercially available

standard tissue DNA is approximately 5%. Furthermore, the minimum detection sensitivity for copy number is 4 copies.

Follow-up

The participants will be evaluated at scheduled visits during screening, treatment, and follow-up. The participants will be followed up every 1 year after the initial surgery. Follow-up is continued until death or a maximum of 3 years after surgery. To promote complete follow-up, the coordinating office will send an e-mail at the follow-up time of each enrolled patient. At any time, patients can withdraw their informed consent regarding the use of biological specimens by requesting the site staff.

Data management

In this study, all patient data are collected by EDC system. User identification code and password are provided from the data manager. Participant data are anonymized using registration number issued by the data manager at the registration. Data cleaning is performed regularly to avoid duplicates, missing data, and inconsistencies.

Monitoring

This study does not require a data monitoring committee (DMC) because it is observational, and there is no need for safety monitoring. Moreover, the sample size is small so an

interim analysis has not been planned. Site visit monitoring and auditing will not be undertaken unless critical problems are confirmed.

Protocol amendment

Any protocol amendment or change which may impact on the conduct of the study will be reviewed and approved by the Ethics Committee prior to its activation. The coordinating office will notify important protocol modifications to relevant parties.

Consent or assent

Physicians provide the patient enough time to contemplate and that the patient has understanding on the study participation. If the patient agrees to participate in the study, the physician will obtain the patient's signature on the ICF reviewed and approved by each site. Informed consent form includes the information about secondary use of patient data and biological samples for future study. Future research will only use data/samples from patients who have consented to secondary use.

Confidentiality

All participant information used during the anonymization process will be stored and managed in a password-protected device. All samples, reports, and data will be anonymized by registration number and stored in an area with limited access. All records containing names or other personal data will be stored in locked file cabinets with limited access.

Access to data

The project principal investigator can access the final cleaned data set, and with the approval of the project principal investigator, other study team members can access anonymized data.

Dissemination policy

Primary publications will be published in English language journals. Conference presentations and articles (review articles) for referral of studies that do not include the results of the analysis of study endpoints, as well as the distribution of patient characteristics and the presentation of safety data after completion of enrollment, can be performed with the approval of the principal investigator.

In principle, the authors of the main published articles on the results of the study (the first publication of the results of the primary endpoint) will be the researchers of each institution, and the following will be finalized by the principal investigator depending on their contributions, including the biostatistician responsible for the analysis and the researchers who have contributed to the study.

All co-authors will review the article content prior to publication and only those who agree to the publication. In principle, the leading presenter of the main conference presentation (the first conference presentation of the results of primary endpoint) will be the principal investigator.

All publications relevant to the study should comply with the ethical standards widely accepted for publication and writing, including the Uniform Requirements for Manuscripts Submitted to Biomedical Journals specified by the International Committee of Medical Journal Editors (Unified Requirements for Manuscripts in Biomedical Journals), the Ethical Considerations in the Conduct and Reporting of Second Section Studies of <http://www.icmje.org/index.html#authorship>.

A publication containing the results of this study has not yet been published or submitted. In addition, no any parts of this study have yet been presented at a conference.

Storage of sample and information

The study information should be stored for 5 years from submission date of the final analysis report or 3 years from study related publication date, whichever comes later, and is recommended the study site to extend the storage period as possible.

Storage period in the data center shall be semipermanent, in consideration of long-term follow-up and/or secondary research use. After the storage period, samples and information related to this study should be destroyed after anonymization at each institution. Correspondence tables or any other data used while in the anonymization process should be stored and managed in a password-protected device.

Discussion

This study aims to establish a novel MRD detection assay for HER2-positive early breast cancer by evaluating the association between MRD and recurrence in HER2-positive early breast cancer. Minimal residual disease detection using ctDNA is a less invasive and repeatable technique. Minimal residual disease landmark analysis after completion of all treatment is useful for identifying low-risk patients for whom overtreatment can be avoided and high-risk patients requiring intensive therapy. Repeated ctDNA testing in long-term surveillance is useful in predicting recurrence and may allow treatment to be initiated before clinical recurrence.¹⁵⁻¹⁸

Adjuvant chemotherapy of HER2-positive early breast cancer patients received NAC is decided by pathological response.⁷ However, the recurrence is experienced even in patients with HER2-positive breast cancer who achieved pCR after dual HER2 blockade.⁸ HER2DX is a one of the tools to predict the prognosis of HER2-positive breast cancer.¹⁴ However, a clinically feasible assay to assess the status of the remaining tumor after treatment is lacking.

Multiple studies have shown that detected ctDNA is associated with prognosis.^{15,16} The I-SPY2 trial in all subtypes of breast cancer suggested that ctDNA levels after NAC, but not pCR, were associated with recurrence.¹⁹ Although the NeoALTTO trial of HER2-positive breast cancer could not detect a significant association between ctDNA and event-free survival, it reported that ctDNA detection at baseline was associated with PAM50 subtypes and lower pCR rate.²⁰ During the follow-up surveillance, ctDNA is identified before clinical evidence of recurrence. Median lead times for ctDNA are reported at 8.9 months for Coombs and 12.4 months for Lipsyc-Sharf. Circulating tumor DNA surveillance tends to improve sensitivity compared with landmark analysis.^{17,18} However, it is unclear whether treatment changes based on ctDNA detection will improve outcomes. Trials to evaluate whether ctDNA-guided intervention alters clinical outcomes are underway.²¹

There are several approaches for ctDNA analysis of MRD, such as digital polymerase chain reaction (PCR), PCR amplicon-based next-generation sequencing (NGS), hybridization capture-based NGS, whole-genome sequencing, and epigenomic approaches (ie, methylation and fragmentomics). However, a standard method has not yet been established because of the lack of the sensitivity and specificity required for clinical application.¹⁵ In this study, we use the TOP and DNA methylation analysis. Today OncoPanel is a hybridization capture-based NGS that can analyze a wide range of cancer somatic gene mutations using nucleic acids obtained from tumor tissues with a DNA panel and simultaneously perform fusion gene detection analysis with an RNA panel for a total of more than 900 genes.²² Currently, we have reported the analysis of ctDNA, cfRNA, circulating tumor cells (CTCs) in advanced cancer using the TOP platform.^{23,24} In this study, we will evaluate the utility of TOP for MRD detection.

From another perspective, there are 2 different approaches: tumor genotype-informed and genotype-naïve methods. Tumor genotype-informed methods have high sensitivity and specificity because they can reduce the influence of genetic aberrations of nontumor origin (ie, CHIP).²⁵ However, patients who achieve pCR with NAC do not have sufficient tumor volumes for genetic analysis. Genotype-naïve methods are useful for detecting genetic aberrations acquired during treatment.²⁶ Therefore, we will evaluate our detection method to determine whether it can be used as a genotype-naïve method. However, we will perform tissue genetic analysis to confirm genotype concordance between the tissue and plasma.

In addition to detection methods, biological features also affect the MRD results. The amount of ctDNA correlates with tumor burden.²⁷ Moreover, the amount of ctDNA released into blood varies depending on the organ involved. Circulating tumor DNA detection rate in patients with brain tumor is less than 10%.²⁸ HER2-positive breast cancer is reported to have a higher ctDNA level than the luminal type.¹¹ In addition,

cancer-specific gene mutations (ie, *EGFR* mutations in lung cancer) are good targets for detecting of MRD. Variants of the *BRCA1*, *BRCA2*, and *PALB2* gene are reported to be associated with a high risk of breast cancer (odds ratio ranging from 5.0 to 10.6), but their frequency is low.²⁹ HER2-positive breast cancer has a high frequency of *ERBB2* amplification and *TP53* mutation.³⁰ We plan to detect ctDNA/ctRNA using a panel including *ERBB2* and *TP53*.

The limitation of this study is small sample size. Although exploratory, this prospective study will provide the basic data for new MRD assay. Studies on MRD assay are important to enable risk-based individualized therapy.

Conclusions

The findings of this study will provide insights into the utility of a new MRD assay and the relationship between landmark/surveillance MRD and recurrence in patients with early HER2-positive breast cancer.

Declarations

Ethics approval and consent to participate

The National Cancer Center Hospital Certified Review Board approved the study protocol in March 2022, and participant enrollment began in June 2022 (Institutional Review Board registry number: 2021-364). All participants provided informed written and informed verbal consent. All study was conducted in accordance with the Declaration of Helsinki.

Consent for publication

Not applicable.

Author contributions

MT involved in conceptualization, investigation, and writing—original draft. **MA involved in** investigation and writing—original draft. **YK involved in** conceptualization, investigation, methodology, and writing—review and editing. **RK involved in** investigation and writing—review and editing. **SY involved in** conceptualization and writing—review and editing. **CA involved in** investigation and writing—review and editing. **RL involved in** investigation and writing—review and editing. **MP involved in** investigation and writing—review and editing. **AG involved in** investigation and writing—review and editing. **MM involved in** investigation and writing—review and editing. **MO involved in** investigation and writing—review and editing. **MI involved in** investigation, supervision, and writing—review and editing. **KS involved in** conceptualization, investigation, and writing—review and editing. **TS involved in** investigation, supervision, and writing—review and editing. **AS involved in** investigation, supervision, and writing—review and editing. **SK involved in** data curation, methodology, and writing—review and editing. **RM involved in** formal analysis and writing—review and editing. **RS involved in** formal analysis and

writing—review and editing. **MY involved in** investigation and writing—review and editing. **YY involved in** investigation and writing—review and editing. **TH involved in** methodology, project administration, and writing—review and editing. **KN involved in** conceptualization, supervision, and writing—review and editing. **KY involved in** investigation, supervision, and writing—review and editing. **SS involved in** conceptualization, investigation, project administration, and writing—review and editing.

Acknowledgements

We thank all the participants for their participation in this study. Editage-Cactus Communications provided editorial support in the form of medical editing, assembling tables, creating high-resolution images based on the authors' detailed directions, collating author comments, copyediting, fact checking, and referencing.

Availability of data and materials

Not applicable.

Trial registration

ClinicalTrials.gov: NCT05433753 (registered 27/06/2022).

Protocol version

Issue date: June 7, 2023, Protocol amendment number: ver1.2, Authors: YK, SS, RK, RM, and RS

Revision chronology

Ver1.0 2022-MarVer1.0-22 Original

Ver1.1 2022-JunVer1.1-6 Amendment 01

Primary reason for amendment: H.U. group Institute G.K. has been added as a joint research institution.

Ver1.2 2023-JunVer1.2-7 Amendment 02

Primary reason for amendment: Changes in research members due to researcher movement. Added the definition of DFS to better clarify handling secondary cancer.

Sponsor contact information

Trial Sponsor: National Cancer Center Hospital

Contact name: Yuki Kojima

Address: 5-1-1 Tsukiji, Chuo-ku, Tokyo, 104-0045, Japan.

Phone: (+81)-3-3542-2511

E-mail: ncch2113_office@ml.res.ncc.go.jp

Role of committees

Principal investigator (sponsor): Design and conduct of this study and member of coordinating office.

Coordinating office: Study planning, organization of committee meetings, preparation of protocol and revisions, recruitment of patients, reviewing the progress of study, and sample collection.

Data manager: Maintenance of EDC system and data verification.

Biostatistician: Statistical data processing.

Pathologists and genome analysts: Pathological evaluation and genomic analysis.

ORCID iD

Momoko Tokura  <https://orcid.org/0009-0008-2344-4057>

REFERENCES

1. Sung H, Ferlay J, Siegel RL, et al. Global Cancer Statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J Clin.* 2021;71:209-249.
2. Heer E, Harper A, Escandor N, et al. Global burden and trends in premenopausal and postmenopausal breast cancer: a population-based study. *Lancet Glob Health.* 2020;8:e1027-e1037.
3. McTiernan A, Porter P, Potter JD. Breast cancer prevention in countries with diverse resources. *Cancer.* 2008;113:2325-2330.
4. Allemani C, Matsuda T, Di Carlo V, et al. CONCORD Working Group. Global surveillance of trends in cancer survival 2000-14 (CONCORD-3): analysis of individual records for 37 513 025 patients diagnosed with one of 18 cancers from 322 population-based registries in 71 countries. *Lancet.* 2018;391:1023-1075.
5. Loibl S, Gianni L. HER2-positive breast cancer. *Lancet.* 2017;389:2415-2429.
6. Slamon D, Eiermann W, Robert N, et al. Adjuvant trastuzumab in HER2-positive breast cancer. *N Engl J Med.* 2011;365:1273-1283.
7. Mamounas EP, Untch M, Mano MS, et al. Adjuvant T-DM1 versus trastuzumab in patients with residual invasive disease after neoadjuvant therapy for HER2-positive breast cancer: subgroup analyses from KATHERINE. *Ann Oncol.* 2021;32:1005-1014.
8. O'Shaughnessy J, Robert N, Annavarapu S, et al. Recurrence rates in patients with HER2+ breast cancer who achieved a pathological complete response after neoadjuvant pertuzumab plus trastuzumab followed by adjuvant trastuzumab: a real-world evidence study. *Breast Cancer Res Treat.* 2021;187:903-913.
9. Riggio AI, Varley KE, Welm AL. The lingering mysteries of metastatic recurrence in breast cancer. *Br J Cancer.* 2021;124:13-26.
10. Parsons HA, Rhoades J, Reed SC, et al. Sensitive detection of minimal residual disease in patients treated for early-stage breast cancer. *Clin Cancer Res.* 2020;26:2556-2564.
11. Spring LM, Fell G, Arfe A, et al. Pathologic complete response after neoadjuvant chemotherapy and impact on breast cancer recurrence and survival: a comprehensive meta-analysis. *Clin Cancer Res.* 2020;26:2838-2848.
12. Sparano JA, Gray RJ, Makower DF, et al. Adjuvant chemotherapy guided by a 21-gene expression assay in breast cancer. *N Engl J Med.* 2018;379:111-121.
13. Kalinsky K, Barlow WE, Gralow JR, et al. 21-gene assay to inform chemotherapy benefit in node-positive breast cancer. *N Engl J Med.* 2021;385:2336-2347.
14. Prat A, Guarneri V, Pascual T, et al. Development and validation of the new HER2DX assay for predicting pathological response and survival outcome in early-stage HER2-positive breast cancer. *EBioMedicine.* 2022;75:103801. doi:10.1016/j.ebiom.2021.103801
15. Moding EJ, Nabet BY, Alizadeh AA, et al. Detecting liquid remnants of solid tumors: circulating tumor DNA minimal residual disease. *Cancer Discov.* 2021;11:2968-2986.
16. Cescon DW, Bratman SV, Chan SM, et al. Circulating tumor DNA and liquid biopsy in oncology. *Nat Cancer.* 2020;1:276-290.
17. Coombes RC, Page K, Salari R, et al. Personalized detection of circulating tumor DNA antedates breast cancer metastatic recurrence. *Clin Cancer Res.* 2019;25:4255-4263.
18. Lipsyc-Sharf M, de Bruin EC, Santos K, et al. Circulating tumor DNA and late recurrence in high-risk hormone receptor-positive, human epidermal growth factor receptor 2-negative breast cancer. *J Clin Oncol.* 2022;40:2408-2419.
19. Magbanua MJM, Swigart LB, Wu HT, et al. Circulating tumor DNA in neoadjuvant-treated breast cancer reflects response and survival. *Ann Oncol.* 2021;32:229-239.
20. Rothé F, Silva MJ, Venet D, et al. Circulating tumor DNA in HER2-amplified breast cancer: a translational research substudy of the NeoALTTO phase III trial. *Clin Cancer Res.* 2019;25:3581-3588.
21. Pfister K, Huesmann S, Friedl TWP, et al. The SURVIVE study: liquid biopsy guided surveillance after intermediate- to high-risk early breast cancer. *J Clin Oncol.* 2024;42:TPS620. doi:10.1200/JCO.2024.42.16_suppl.TPS620

22. Kohsaka S, Tatsuno K, Ueno T, et al. Comprehensive assay for the molecular profiling of cancer by target enrichment from formalin-fixed paraffin-embedded specimens. *Cancer Sci.* 2019;110:1464-1479.
23. Koyama T, Shimizu T, Kojima Y, et al. Clinical Activity and Exploratory Resistance Mechanism of Milademetan, an MDM2 Inhibitor, in Intimal Sarcoma with MDM2 Amplification: An Open-Label Phase Ib/II Study. *Cancer Discov.* 2023;13:1814-1825.
24. Hasegawa N, Kohsaka S, Kurokawa K, et al. Highly sensitive fusion detection using plasma cell-free RNA in non-small-cell lung cancers. *Cancer Sci.* 2021;112:4393-4403.
25. Abbosh C, Swanton C, Birkbak NJ. Clonal haematopoiesis: a source of biological noise in cell-free DNA analyses. *Ann Oncol.* 2019;30:358-359.
26. Aggarwal C, Thompson JC, Black TA, et al. Clinical implications of plasma-based genotyping with the delivery of personalized therapy in metastatic non-small cell lung cancer. *JAMA Oncol.* 2019;5:173-180.
27. Yi Z, Ma F, Rong G, et al. The molecular tumor burden index as a response evaluation criterion in breast cancer. *Signal Transduct Target Ther.* 2021;6:251. doi:10.1038/s41392-021-00662-9
28. Bettgowda C, Sausen M, Leary RJ, et al. Detection of circulating tumor DNA in early- and late-stage human malignancies. *Sci Transl Med.* 2014;6:224ra24. doi:10.1126/scitranslmed.3007094
29. Breast Cancer Association Consortium; Dorling L, Carvalho S, Allen J, et al. Breast cancer risk genes—association analysis in more than 113,000 women. *N Engl J Med.* 2021;384:428-439.
30. The Cancer Genome Atlas Network. Comprehensive molecular portraits of human breast tumours. *Nature.* 2012;490:61-70.

Appendix

Sample patient informed consent form

Name of Research Study: Asian multicenter prospective study in HER2-positive early breast cancer for detecting minimal residual disease by circulating tumor DNA analysis with neo-adjuvant chemotherapy; HARMONY study

Protocol Number: NCCH2113/HARMONY

Name of Facility Sponsoring the Research Study: National Cancer Center Hospital, Japan

Name of principal investigator (study doctor):

Address of Research Site:

Phone Number:

1. Introduction

You are invited to take part in a “Asian multicenter prospective study in HER2 (human epidermal growth factor receptor 2)-positive early breast cancer for detecting minimal residual disease by circulating tumor DNA analysis with neoadjuvant (presurgical) chemotherapy; HARMONY study.” This research study is being conducted by National Cancer Center Hospital, Japan (also called “NCCH”).

Treatments and examinations you can access nowadays are based on clinical studies. These studies are mostly based on patient cooperation and volunteering.

We are conducting this research to improve the treatment of breast cancer patients with the same condition as yours. This proposal has been reviewed and approved by Medical Research Ethics Committee, which is a committee whose task is to make sure that research participants are protected from any harm. This form contains information that will help you decide whether to join the study. It is important that you understand why the research is being done and what it will involve. Taking part in this research project is voluntary. You do not have to participate whether you do not want to and you can withdraw at any time.

2. Right to refuse or withdraw

You do not have to take part in this research if you do not wish to do so. You may also stop participating in the research at any time. Your decision will not affect your medical care. Refusal to participate will involve no other penalties or loss of benefits to which you are entitled. If you withdraw or are removed from the study, any remaining biological samples (eg, blood samples) that have been collected from you can be destroyed by making a request to the study doctor. However, any data already generated from your samples will be kept to preserve the value of the study.

The following sections describe the details of the research study. Before you decide to take part and sign the consent form, please take as much time as you need to ask questions with the study team (your study doctor and staff), with family and friends, or with your personal physician or other health care professionals. The study team will fully answer any questions you have before you make a decision.

3. Participant selection

Investigators are inviting patients aged 20 years and above, with a diagnosis of HER2-positive breast cancer scheduled to receive presurgical chemotherapy and surgery. The recurrence risk of HER2-positive breast cancer is high, because HER2 is a cell surface protein that relates to cancer cell proliferation. However, anti-HER2 drugs have dramatically prolonged survival. In addition, to reduce the recurrence of HER2-positive breast cancer, the postsurgical chemotherapy is modified based on the presence or absence of cancer in surgical specimens in the patients receiving presurgical chemotherapy. However, there are no effective biomarkers to predict prognosis in patients after presurgical chemotherapy.

4. Purpose of this study

The purpose of this study is to clarify whether MRD is associated with prognosis (the course of disease) in patients with HER2-positive breast cancer. Minimal residual disease is a small number of cancer cells that remain in the body during and after treatment. It is believed that these cells may cause recurrence and that people with MRD may be at increased risk of recurrence.

DNA and RNA are molecule that contains the biological information of your own. The fragment of DNA and RNA circulates in the bloodstream (cell-free DNA/cell-free RNA), which is released by the human cells. The fragment of cancer cell-derived DNA/RNA circulates in the bloodstream of cancer patients, circulating tumor DNA/RNA, ctDNA/ctRNA. The presence of MRD could be evaluated by measuring ctDNA/ctRNA in patients who have undergone therapy to completely cure their cancer. However, the utility of ctDNA/ctRNA testing for MRD detection in Asian cancer patients has not been fully evaluated.

Although the number of breast cancer patients in Asia is increasing, the research on Asian breast cancer patients is

insufficient. It has also been reported that the profiles differ between breast cancer patients in Europe and the United States and those in Asia. Research is needed to improve the prognosis of Asian breast cancer patients. If this study can show the usefulness of ctDNA/ctRNA testing for predicting recurrence, it enables patients to receive the treatment according to their recurrence risk in the future.

5. Number of participants and duration of this study

There will be about 20 patients enrolled from the Philippines in this study. The study is being conducted at about 2 different research sites in the Philippines and Japan. This study will start from the day it was approved by the Independent Ethics Committee and will continue for at least 2 years until March 2024.

The end date may be extended according to future protocol revisions.

Expected enrollment period: 2 years (from April 2022 to March 2024)

Follow-up period: 3 years after completion of enrollment

Expected study period: 6.5 years (from April 2022 to September 2028)

6. Method of this study

(What will happen if I decide to take part?)

(1) Genetic analysis of blood samples

If you participate in this study, we will collect blood from you by drawing about 20 mL (collected at a time), which is about 4 teaspoons of blood from a vein in your arm. Your blood will be collected max 6 times (total 120 mL; 20 mL × 6 times): before presurgical chemotherapy, after presurgical chemotherapy, after surgery, and every year from surgery to 3 years later. If recurrence happened, blood collection would be performed at that time and no further blood samples will be collected. We may ask your permission to collect extra blood samples if the genetic test is not successful. Additional blood collection beyond clinical practice is not required as the collection is performed at the time of your normal clinic laboratory procedure. These blood samples will be sent to the National Cancer Center in Japan.

Genes (DNA/RNA) will be extracted from the submitted blood, and genetic analysis will be performed. Analysis of blood samples and information will be stored by the National Cancer Center Japan.

(2) Genetic analysis of tumor tissue

Genetic analysis of your tumor tissues will be performed to examine whether any genetic alteration found in the tumor tissue is also detected in the blood sample collected. Tumor tissues will be analyzed using samples that have been stored as residual specimens at your hospital after surgery or biopsy.

Genes (DNA/RNA) will be extracted from the submitted tissues, and genetic analysis will be performed. Analysis will be performed, and information will be stored in the National Cancer Center Japan.

(3) Other issues

Congenital individual differences in genes that make various body proteins (called “genetic polymorphisms” or “germline genes”) may be analyzed to examine their effects on the development of cancer and response to drug therapy.

Tumor tissues may be examined for protein substances using methods such as immunohistochemical staining.

We will collect your clinical information from your medical records and analyze ctDNA detection and clinical information to explore whether the presence or absence of ctDNA is a predictor of prognosis or correlates with the effects of presurgical chemotherapy.

7. Information about study risks and benefits

(1) Risks associated with the study

The chance that you will be physically injured as a result of participating in this study is small. In total, 20 mL (total 120 mL, 6 points) of your blood will be collected at a time. You may experience mild pain, bleeding, bruising, and infection at the site of the needle insertion when a sample is drawn from your blood. Some people may faint or feel light-headed, but this usually lasts only a few minutes.

(2) Expected benefit

You may not receive any personal benefits from being in this study. However, others may benefit from the knowledge gained from this study in the future. If this study shows that measurement of ctDNA using blood is useful as a prognostic biomarker, the presence or absence of ctDNA may make decisions about adding or omitting postsurgical therapy after surgery in patients with cancer in the future, thus allowing optimal treatment selection for each patient.

8. What will happen if I do not participate

Your treatment plan will not be affected by your decision. Refusal of participation will involve no penalties or loss of benefits.

9. Cost

You will not incur any additional expenses from participating in this study. Information from this study, including information from research on your samples, may lead to discoveries, inventions, or the development of commercial products. You and your family will not receive any benefits or payment if this happens.

10. Compensation

The chance that you will be physically injured as a result of participating in this study is very small. However, if you are physically injured as a result of participating in this study, the medical treatment cost for your research-related injury will not be borne by the sponsor. Each study site shall provide appropriate treatment for the patient's health damage and daily medical practice.

11. Protection of personal information

The information, such as age, date of birth, medical information records, and results of study tests, will be recorded by the study team and reported to NCCH. Those data will be kept by the NCCH central research database. However, your name and contact information will not be put into the database, and any information that is released will not identify you.

The samples received will be transferred to the analysis institute (National Cancer Center Japan or a Japanese laboratory [Takara Bio Inc.]) under anonymized conditions, and only the anonymization number will be used for sample analysis. Your personal information will not be used in a way that you can be identified.

In addition, to confirm the properness of study conduct and the accuracy of the recorded information, documents such as your medical record may be examined by NCCH or its contracted organization to perform development work; members of the independent ethics committee (IEC) who have examined and reviewed this study; and regulatory authority. National Cancer Center Hospital will obtain confidential information from you, such as past and current diseases, and share them with the parties above. Some of those parties may be outside your home country.

Those parties above will all be obligated to protect your confidentiality, and NCCH will take all necessary precautions to protect your personal information so that your personal information will not be revealed to any other party. In accordance with the applicable laws and regulations, your personal information will be protected.

12. Samples

The collected samples and the research data including genetic testing and clinical information will be stored in the NCCH and St. Luke's medical center at least 5 years after the completion of this study, or as long as possible, for future analysis. All data are anonymized strictly. After the storage period has passed, we will discard the data or information by the appropriate destruction procedures.

Currently, no further studies are planned using the samples or any other information for this study including genomic data. However, there is the possibility of doing a further study using the data of this study if the scientific and ethical aspects of the new study are fully reviewed and approved by the IEC.

13. Additional use of your samples and data— OPTIONAL:

- You can also choose to allow your residual blood sample and tumor tissue or data to be used in future research after study completion for the following topics:
 - > In investigations of the disease under study in the trial and related conditions.
 - > In comparisons with information from the samples of other people, including subjects with other conditions or diseases. This is called using the sample as a “control.” This also includes using the samples to study natural variation in DNA, RNA (ribonucleic acid), proteins, or metabolites, or to develop new pharmacogenomic or biomarker technologies.
- If you do not want your residual tumor to be used in this additional research, you can still take part in this study. You will be asked whether or not you agree to future use for the additional research on your residual tumor at the end of this form.
- The analyzed data may be registered with public databases such as NBDC (<https://biosciencedbc.jp/>) in a form that does not identify individuals.

14. Additional information about this study

A description of this study will be available on the website (<http://www.ClinicalTrials.gov>). The website will not include information that can identify you. At most, the website will include a summary of the results. You can refer it at any time. The website above is in English. If you need assistance understanding the content on the website, please ask your study doctor.

15. Sharing the results

The clinical trial results may be published in medical journals or at a scientific congress or used by regulatory authorities to assess the data by themselves in a different way. In such cases, personal information that could be used to identify you will be kept strictly confidential. Thus, your privacy will be protected. After you participate in the study, even if you discontinue participation, the information collected up to the point of discontinuing will be handled similarly.

16. Disclosure of genetic analysis results and genetic counseling

Results of genetic analysis of patient samples will be collected in the study. The implications of most genetic alterations evaluated in this study have not been established. Thus, we will not disclose the results to patients. If you have concerns about your genetic information, please ask your attending doctor.

17. Conflict of interest

A conflict of interest (COI) is a situation in which a person or organization is involved in multiple interests, financial or otherwise, one of which could possibly corrupt the motivation or decision-making of that individual or organization.

National Cancer Center Hospital is sponsoring the study and will be financing (study doctor/institution) to conduct the study. We have no conflicts of interest directly relevant to the content of this study. We will inform you on the website of NCCCH when there are any changes. If you have any questions about potential conflicts of interest, please ask your study doctor.

18. Financing

This project is funded by Japan Agency for Medical Research and Development (AMED). In addition, part of this research (processing of blood samples collected in Japan) will be conducted under a joint research agreement between the National Cancer Center and the H.U. Group Research Institute G.K., a limited liability company, and the H.U. Group Research Institute G.K. will bear the costs associated with specimen processing. We will perform research while ensuring the transparency and reliability of the research so as not to affect the research results.

Joint Research Institution

Hiroko Higashimoto

H.U. Group Research Institute G.K.

H.U. Bioness Complex, Fuchigami 50

Akiruno City, Tokyo 197-0833, Japan

19. Contact information

If you have any concerns or complaints about this study or how it is being managed, please do not hesitate to discuss your concerns with the study team. The phone numbers to reach the study team are on the first page of this document. If you do not feel comfortable discussing your complaint with the study team, please contact the IEC, Patient Rights Advocate, Institutional Contact, or Bioethicist listed below.

If you have any questions about your rights as a research participant, or you would like to obtain information or offer input, or you wish to speak with someone **not** directly involved with the study, you should contact:

Provide the name, phone number, and address of any of the following: (1) IEC; (2) patient rights advocate; (3) institutional contact; and/or (4) bioethicist; (5) mobile phone numbers of each of the study team members.

CONSENT TO TAKE PART IN A RESEARCH STUDY

Name of Institution:

Name of Research Study: Asian multicenter prospective study in HER2-positive early breast cancer for detecting MRD by

circulating tumor DNA analysis with neoadjuvant chemotherapy; HARMONY study

1. Introduction
2. Right to refuse or withdraw
3. Participant selection
4. Purpose of the study
5. Number of participants and duration of this study
6. Method of this study
7. Information about study risks and benefits
8. What will happen if I would not participate
9. Cost
10. Compensation
11. Protection of personal information
12. Samples
13. Additional use of your samples and data
14. Additional information about this study
15. Sharing the results
16. Disclosure of genetic analysis results and genetic counseling
17. Conflict of interest
18. Financing
19. Contact information

Consent to participate in the research study

By signing this document, you are agreeing to be in this study. Please make sure you understand what the study is about before you sign. We will give you a copy of this document for your records, and we will keep a copy with the study records. If you have any questions about the study after you sign this document, you can contact the study team.

I understand the content of this clinical study and its risks and benefits and my questions so far have been answered. I consent to take part in this study.

OPTIONAL:

RESIDUAL BLOOD SAMPLE AND TUMOR TISSUE TO BE PLACED INTO STORAGE FOR FUTURE CANCER RESEARCH

You can indicate your choice to store your sample for future research nor not by checking the box below:

- ☐ I agree to have the remaining sample placed into storage for future cancer research for developing tests to accurately determine who will gain the most benefit, after study completed.

OR

- ☐ I **do not agree** to have any remaining sample placed into storage for future cancer research.

Please destroy my sample when you are certain it will no longer be required in this study.

PATIENT CONSENT

Signature of study participant Date of signature

AUTHORIZED PERSON OBTAINING CONSENT

Signature of Legally Authorized Representative ☐☐☐ Date of signature §

Signature of the Person Conducting the Date of signature
Consent Discussion †

Signature of impartial witness ‡ Date of signature §

§ Subject/legal representative/impartial witness must personally date their signature.

† The investigator, or an appropriately qualified and trained person designated by the investigator to conduct the informed consent process, must sign and date the consent document during the same interview when the subject signs the consent document.

‡ Impartial Witness: A person, who is independent of the study, who cannot be unfairly influenced by people involved with the study, who attends the informed consent process, and who reads the informed consent and any other written information supplied to the subject. Guidance for Industry E6 Good Clinical Practice: Consolidated Guidance.

For countries that require a separate consent page along with the information sheet (consent document), the following statement of consent form can be utilized.

CONSENT FORM

AGREEMENT TO PARTICIPATE

YOUR CONSENT	(IF REQUIRED, STUDY PARTICIPANT INITIALS)
1. I confirm I have read and understand the information sheet dated <enter date of information sheet> for the above study and have had the opportunity to ask questions. I have been given enough time and opportunity to ask about the details of the study and to decide whether or not to participate in the study. I am in the <enter age range> age range.	
2. I understand that my participation is voluntary and that I am free to withdraw without giving any reason, without my medical care or legal rights being affected.	
3. I understand that others working on NCCH's behalf, ethics committees or institutional review boards, and regulatory agencies and bodies will need my permission to look at my health records in respect of this study and any further research, and I agree to this access.	
4. I consent to the collection, processing, reporting and transfer of my health information within and outside my country of residence for health care and/or medical research purposes as described in the information sheet.	
5. I agree not to restrict the use of any data or results, which arise from this study.	
6. I agree to take part in the above study.	